SECTION C: ARQ PROTOCOL SCENARIOS & TIMEOUTS (10 MARKS)

Q6. ARQ Protocol Operation *[5 Marks]*

The diagram below outlines an Automatic Repeat Request (ARQ) transmission sequence between Computer A and Computer B:

```
Step 1: Computer A sends Packet 1 → starts Timeout Timer.
Step 2: Computer B receives Packet 1 → calculates Checksum → Checksum matches.
Step 3: Computer B sends [ Message X ] → Computer A cancels timer → sends Packet 2.
Step 4: Packet 2 gets corrupted by electrical noise in transit.
Step 5: Computer B detects checksum mismatch → sends [ Message Y ].
Step 6: Computer A receives Message Y → immediately re-transmits [ Packet Z ].
```

(a) Identify Message X: _____ *[1]*

(b) Identify Message Y: _____ *[1]*

(c) Identify Packet Z: _____ *[1]*

(d) Explain what Computer A will do if Packet 1 was completely dropped by a faulty router and Computer B received nothing: *[2]*

Q7. ARQ Timeout Timer Evaluation *[5 Marks]*

(a) Explain what would happen if the sender's timeout timer was configured to be **too short**: *[2.5]*

(b) Explain what would happen if the sender's timeout timer was configured to be **too long**: *[2.5]*